ARCHITECTURE AND DECISION RECORD

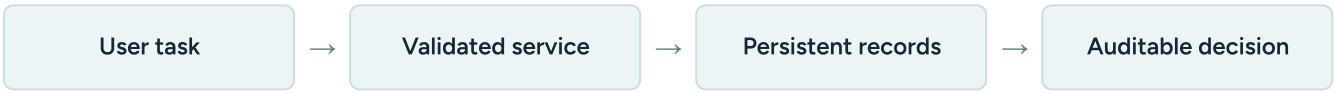
An offline community information application

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Flutter | Dart | Firebase | Offline sync

Problem and contribution

This original design case study explores how to provide accessible preparedness information while making freshness, connectivity limits, and the need to follow official instructions unmistakable. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- Use versioned content snapshots so a partial sync cannot mix new site names with old instructions.
- Avoid collecting precise location history; an optional current location should not become a permanent tracking record.
- Offer text directions and site contact details, but do not claim an automatically generated route is safe during a disaster.

Core records

Record	Responsibility
guidance_packs	One area has multiple immutable pack versions; a device activates one completed download.
evacuation_sites	A site appears in many pack versions through reviewed content snapshots.
advisories	Advisories require review and expiry metadata; they are not model-generated forecasts.
content_reviews	Each publication links to an accountable content review.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Offline availability	Enable airplane mode after a full download and reopen all core screens.	Downloaded checklists, site details, and update labels remain available.
Interrupted update	Cut connectivity midway through a replacement pack download.	The previous verified pack remains active until the new one is complete.
Accessibility	Use enlarged text, a screen reader, and map-free navigation to find a site.	Essential details can be read in a logical order without relying on color or a pin.
Freshness comprehension	Present volunteers with current, expired, and offline advisory screens.	Record whether users correctly identify the status and intended next action.

Technical review

Keep the last complete offline pack until a replacement is verified. Provide a list alternative to maps, and show when content was reviewed so users do not interpret cached details as live status.

How to evaluate the design

Measure successful retrieval of a named site, comprehension of freshness labels, offline task completion, and storage size per content pack. Record device type and accessibility settings. A usability result is not evidence that a route is safe or that disaster outcomes improve.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No emergency dispatch, live hazard prediction, emergency medical advice, guaranteed route safety, or replacement of official announcements.

Review whether a user can tell the difference between preparedness guidance, an old cached advisory, and a current official instruction.

> **Authorship and reuse**
>
> Prepared by Isabella Reyes as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.